

LAB TASK SUBMISSION

BECE204L-Microprocessor and Microcontroller

TASK-3

Follow the LAB INSTRUCTION template for all the experiments (Refer LAB INSTRUCTION file in msteams)

Question 1:

- a) Generate a square wave using 8051 operating at 40MHz
- b) Generate a rectangular wave using 8051 code operating at 40MHz

Question 2:

- a) Generate a sawtooth wave using 8051 code
- b) Generate a triangular wave using 8051 code operating at 40MHz

Question 3:

Write a program using timer 1 to generate a 1 KHz square wave frequency on one of the pins of P1. Then examine the frequency using the oscilloscope.

Question 4:

Write a program using timer 1 to generate a 2 KHz square wave frequency on one of the pins of P1.0. Then examine the frequency using the KEIL IDE inbuilt Logic Analyzer.

Question 5:

Write a 8051 alp program to transfer letter “A” serially with 4800 baud and do this continuously.

Question 6:

Write a 8051 alp program to transfer letter “HELLO” serially with 9600 baud, 8 bit data, 1 stop bit, and do this continuously.

NOTE:

Supporting materials are uploaded in Msteams/VTOP

For late submission marks will be reduced

Submit before the dead line— **March 12, 2026**